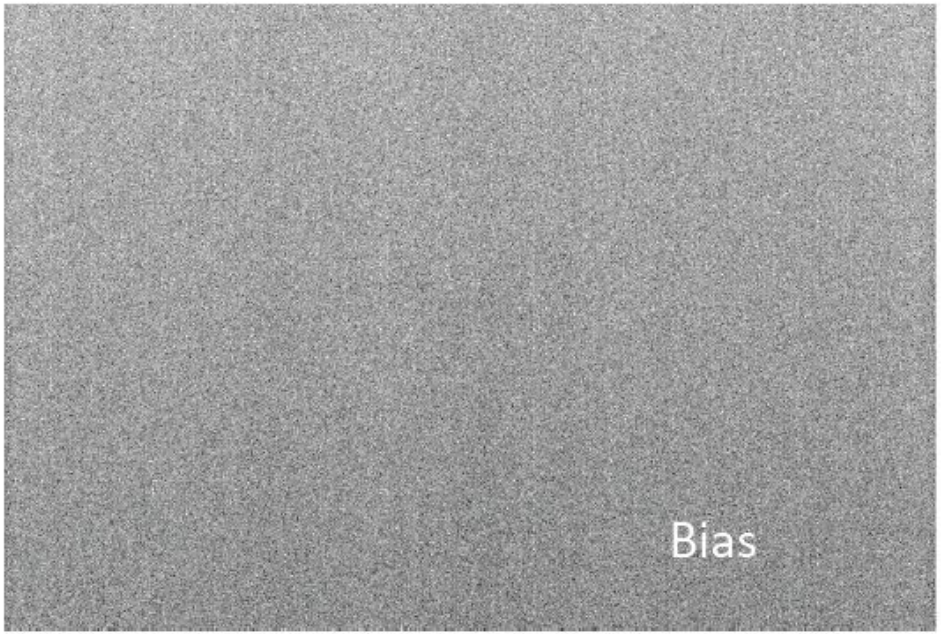
Data reduction

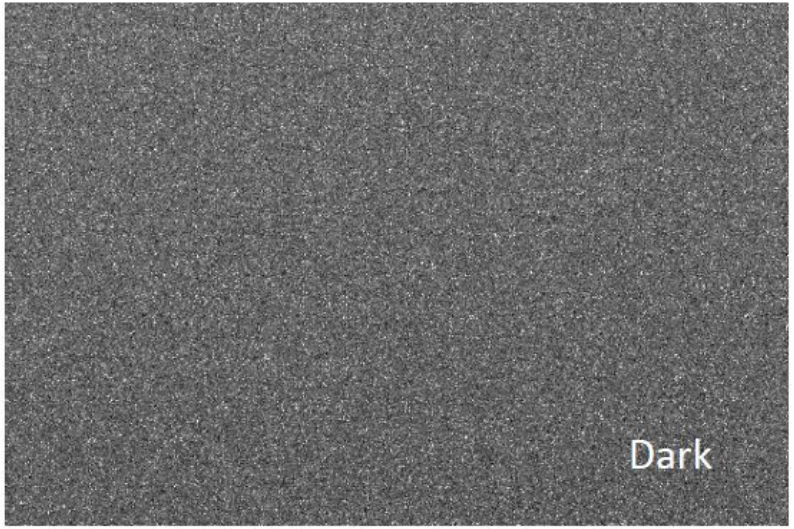
Bias

- A **bias frame** is an image with zero exposure time;
- It’s used to obtain the signal produced by the electronics themselves, the readout noise.



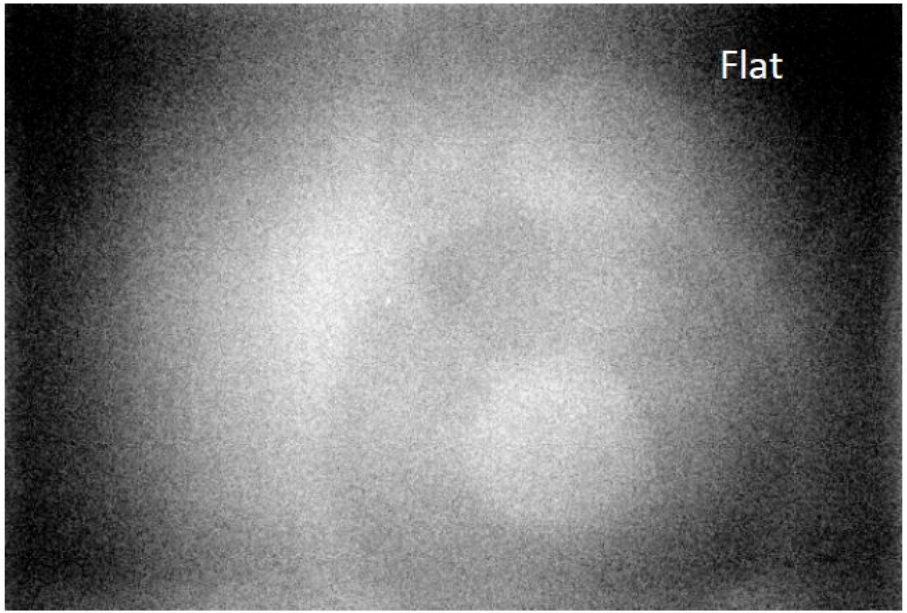
Dark

- A **dark frame** is similar to a bias (no light), but with exposure time (ideally the same exposure as the light frame)
- It’s used to read the signal from thermal fluctuations, the dark current



Flat

- A **flat field** is an image of a constant illumination through the CCD.
- It’s used to measure each pixel’s response to light:
- Flats must be taken in all filters.



Summary

- Bias: ~~Light~~ ~~Exposure Time~~ ~~Filter~~
- Dark: ~~Light~~ Exposure Time ~~Filter~~
- Flat: Light Exposure Time Filter

Stacking

- To increase SNR, we increase exposure time; However, things can saturate.

- **Solution:** We stack many images of shorter exposures: blend several images of the same target together.
- **How:** create a data cube with all the frames and median (average) in the 3rd dimension.
  - Final image is not brighter (signal is the same)!
  - Final image is smoother because it has less noise (noise is random and cancels out)

Data reduction

